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Computer Architecture - Software

Assignment - 3

The benefits of serverless architecture :

Serverless architecture is a new method to application development that allows developers to focus solely on creating code rather than maintaining the underlying infrastructure. Code is executed and scaled automatically using platforms such as AWS Lambda, Azure functions, and Google Cloud Functions. This concept saves operational overhead, cuts expenses by charging only for real consumption, and boosts scalability. It is particularly useful for applications with unpredictable traffic or event-driven workloads.

The concept of progressive Web Apps (PWAs):

Progressive Web Apps (PWAs) are web applications that provide an app-like experience in the browser. They integrate online accessibility with standard native software application capabilities like offline access, push notifications, and home screen installation. PWAs are responsive, fast, and capable of working across all devices and platforms, making them perfect for business aiming to reach a large audience without the complexity of developing separate native applications for Android and iOS.

The role of AI and Machine Learning in Software Architecture

- Artificial intelligence (AI) and machine learning (ML) transform software architecture by enabling systems to learn from data and make intelligent decisions. Instead of depending simply on static rules, applications can now adapt content, personalize it, recognize anomalies, and automate complex tasks. AI/ML models often integrated as microservices or via APIs, ensuring modular and scalable design. Their inclusion in software architecture drives innovations in areas like recommendation systems, fraud detection, chatbots, and predictive analysis.

Cloud computing service models (SaaS, PaaS, IaaS) and their use cases :

Cloud computing provides three main service models, each responding to a distinct requirements. Software as a Service (SaaS) delivers complete applications like Gmail or Microsoft 365 over the internet.

Platform as a Service (PaaS) is a development environment (e.g., Google App Engine, Heroku) that allows developers to create and deploy apps without having to manage infrastructure.

Infrastructure as a Service (IaaS) offers virtualized computing resources like storage, server, and networking through providers such as AWS EC2 or Microsoft Azure. These models provide flexibility, scalability, and cost-effectiveness for a variety of commercial applications.